FREQUENCY RESPONSE AND STABILITY ANALYSIS

Using the Bode Plot Method

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1. Abstract

This report presents the stability analysis of a linear time-invariant (LTI) system using the frequency response approach, specifically the Bode plot method. The concepts of frequency response, magnitude and phase plots, gain margin, and phase margin are discussed in detail. A given transfer function is analyzed both theoretically and numerically to determine system stability. MATLAB is used to validate the analytical results. The study highlights how frequency-domain techniques provide insight into system robustness and performance.

2. Introduction

Stability is a fundamental requirement in control systems. A system is stable if its output remains bounded for any bounded input.

Traditional time-domain methods include:

- Routh-Hurwitz Criterion
- Root Locus

Modern engineering systems — particularly in control, robotics, and electronics — are better analyzed using frequency response methods, where system behaviour is studied as a function of input frequency.

3. Frequency Response

Frequency response describes how a system reacts to sinusoidal inputs of varying frequency.

For a sinusoidal input:

$$r(t) = A \sin(\omega t)$$

The corresponding steady-state output is:

$$c(t) = B \sin(\omega t + \varphi)$$

From these expressions:

- Amplitude ratio = B / A
- Phase shift = φ

The frequency response of the system is represented by:

$$G(j\omega)$$

4. Bode Plot

A Bode plot is a graphical representation of the frequency response and consists of two plots:

4.1 Magnitude Plot

$$20 \log_{10} |G(j\omega)|$$

4.2 Phase Plot

$$\angle G(j\omega)$$

Key features of the Bode plot:

- Logarithmic frequency scale
- Straight-line approximations are possible
- Suitable for high-order systems

5. Advantages of the Bode Plot

- Simplifies multiplication into addition (log scale)
- Easy to sketch manually
- Provides gain and phase stability margins
- Widely used in control system design

6. Standard Procedure for Drawing a Bode Plot

- Convert the transfer function: substitute $s = j\omega$
- Express in standard (time-constant) form
- Identify poles and zeros
- Determine break (corner) frequencies
- Plot magnitude using asymptotic slopes
- Plot phase using individual phase contributions

7. Key Parameters

7.1 Gain Crossover Frequency (ω^{Gc})

The frequency at which the open-loop magnitude equals unity:

$$|G(j\omega)| = 1$$

7.2 Phase Crossover Frequency (ω_p^c)

The frequency at which the phase angle equals −180°:

$$\angle G(j\omega) = -180°$$

7.3 Gain Margin (GM)

The gain margin is the reciprocal of the magnitude at the phase crossover frequency:

$$GM = 1 / |G(j\omega_p^c)|$$

Expressed in decibels:

$$GM\ (dB) = 20 \log_{10}(GM)$$

7.4 Phase Margin (PM)

The phase margin is the additional phase lag required to bring the system to the verge of instability:

$$PM = 180° + \angle G(j\omega^{Gc})$$

8. Stability Criteria

Condition	Stability
GM > 0 and PM > 0	Stable
GM = 0 or PM = 0	Marginally Stable
GM < 0 or PM < 0	Unstable

9. System Under Study

The transfer function of the system being analyzed is:

$$G(s) = 40 / [\ s(s + 2)(s + 20)\ ]$$

10. Frequency Domain Representation

Substituting s = jω:

$$G(j\omega) = 40 / [\ j\omega(j\omega + 2)(j\omega + 20)\ ]$$

10.1 Magnitude Expression

$$|G(j\omega)| = 40 / [\ \omega \times \sqrt{(\omega^2 + 4)} \times \sqrt{(\omega^2 + 400)}\ ]$$

10.2 Phase Expression

$$\angle G(j\omega) = -90° - \tan^{-1}(\omega/2) - \tan^{-1}(\omega/20)$$

11. Numerical Analysis

11.1 Gain Crossover Frequency

Trial calculations are performed to locate ω^{Gc}:

- At $\omega = 0.8$ rad/s $\rightarrow$ $|G| \approx 1.165$
- At $\omega = 1.0$ rad/s $\rightarrow$ $|G| \approx 0.893$

By interpolation:

$$\omega^{Gc} \approx 0.9 \text{ rad/s}$$

11.2 Phase Margin

Evaluating the phase at $\omega = 0.9$ rad/s:

$$\angle G(j\omega) = -116.8°$$

$$PM = 180° - 116.8° = 63.2°$$

11.3 Phase Crossover Frequency

The phase crossover frequency is found by solving:

$$\tan^{-1}(\omega/2) + \tan^{-1}(\omega/20) = 90°$$

$$\omega_p^c \approx 6.3 \text{ rad/s}$$

11.4 Gain Margin

Computing the magnitude at $\omega = 6.3$ rad/s:

$$|G(j\omega)| \approx 0.0458$$

$$GM = 20 \log_{10}(21.83) \approx 26.8 \text{ dB}$$

12. MATLAB Implementation

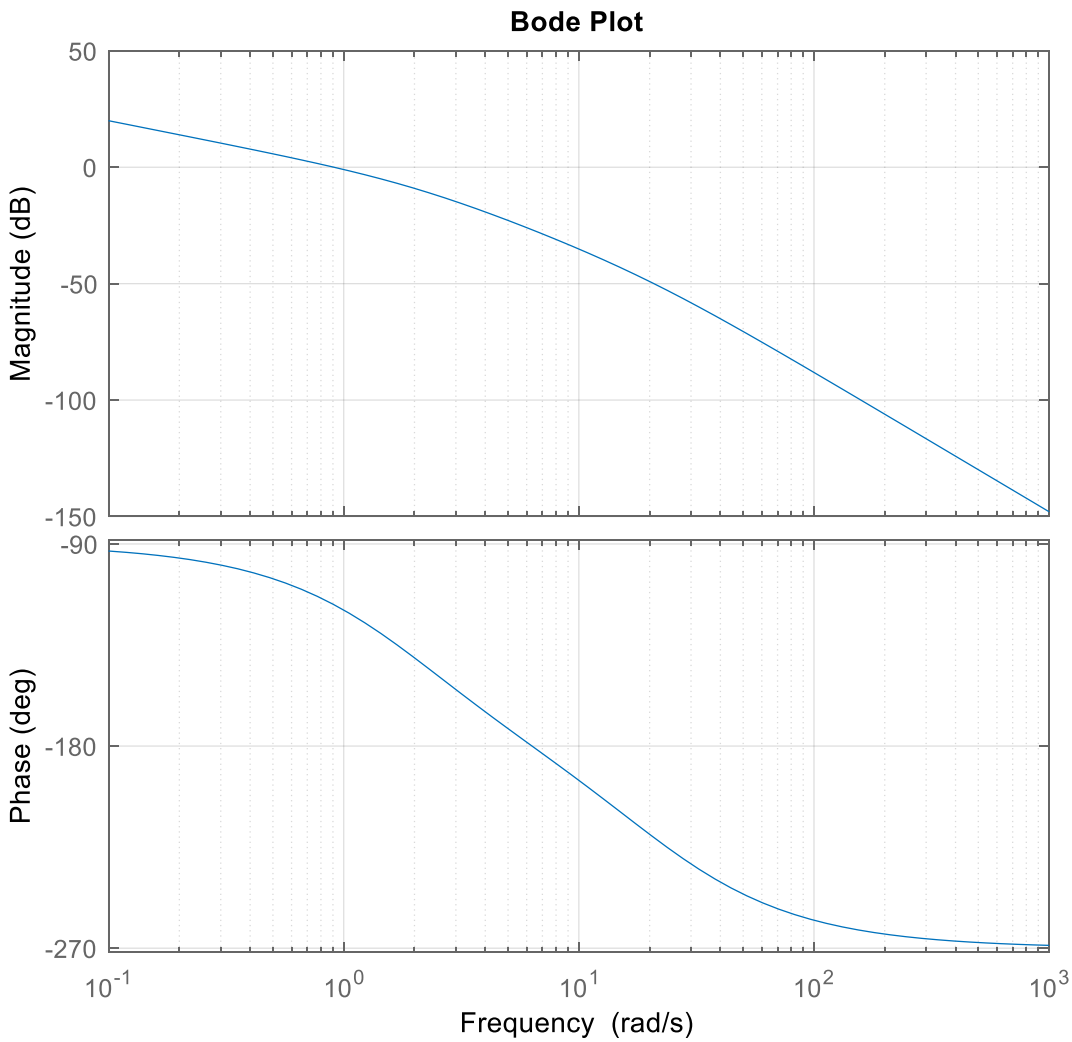
The following MATLAB code generates the Bode plot and computes the stability margins:

```
clc;
clear;
close all;

num = [40];
den = conv([1 0], conv([1 2], [1 20]));

G = tf(num, den);

figure;
bode(G);
grid on;
title('Bode Plot');

[gm, pm, wcg, wcp] = margin(G);
```

```
disp(['Gain Margin (dB): ', num2str(20*log10(gm))]);
disp(['Phase Margin (deg): ', num2str(pm)]);
disp(['Gain crossover freq: ', num2str(wcg)]);
disp(['Phase crossover freq: ', num2str(wcp)]);
```

13. Results

Parameter	Value
Gain Crossover Frequency (ω^{Gc})	≈ 6.32 rad/s
Phase Crossover Frequency ($\omega_p{}^c$)	≈ 0.90 rad/s
Phase Margin (PM)	≈ 62.94°
Gain Margin (GM)	≈ 26.84 dB



14. Discussion

The results indicate the following:

- A phase margin of 63° indicates a well-damped system.
- A gain margin of 27 dB indicates strong robustness against gain variations.
- The system can tolerate significant gain changes without becoming unstable.

In practical terms:

- High PM → smooth transient response
- Low PM → oscillatory response
- High GM → tolerance to parameter uncertainty

15. Conclusion

The Bode plot method provides an effective way to analyze stability in the frequency domain. By evaluating the gain and phase margins, the stability and robustness of the system can be determined. The given system is stable, with a phase margin of 63° and a gain margin of 27 dB, confirming good damping characteristics and high tolerance to gain variations.